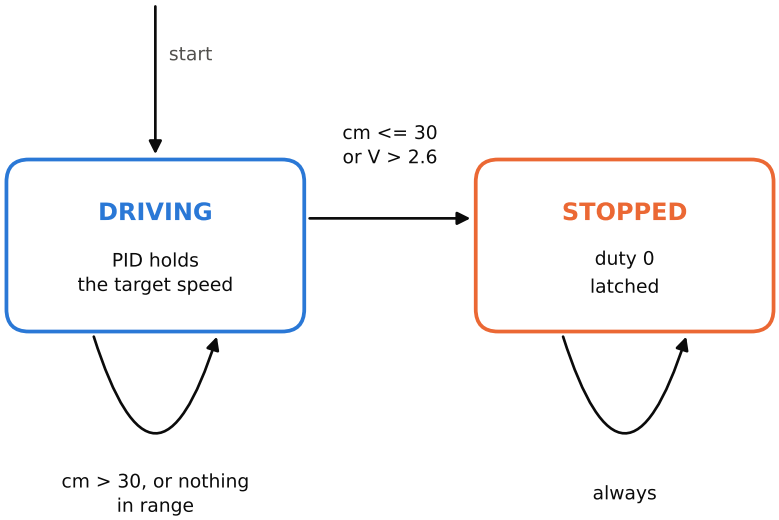

Fig. 9.1. Stop logic of avoid.py, should_stop(), rev d22b77d. Checked once per control loop, about every 82 ms. DRIVING holds the target speed with the Lab 7 PID. Any reading at or under 30 cm, or above 2.6 V (too close to measure), moves it to STOPPED, which is latched until the run ends. The e-stop button is separate and cuts the motor driver in hardware from either state.

t s	volts	cm	stopped	left rpm	left duty	right rpm	right duty
0.496	1.356	37.8	0	69.1	0.0885	68.0	0.0943
0.583	1.408	36.3	0	67.9	0.0952	67.6	0.0994
0.671	1.458	35.0	0	66.3	0.103	68.1	0.1027
0.757	1.516	33.6	0	70.2	0.0987	68.3	0.1061
0.847	1.581	32.2	0	70.2	0.1021	71.4	0.103
0.935	1.648	30.8	0	69.2	0.108	71.7	0.1054
1.044	**1.718**	**29.5**	**1**	**70.3**	**0.0**	**72.6**	**0.0**
1.128	1.819	27.8	1	60.3	0.0	55.0	0.0
1.21	1.858	27.2	1	40.0	0.0	38.2	0.0
1.286	1.881	26.9	1	20.7	0.0	18.9	0.0
1.356	1.919	26.3	1	4.8	0.0	3.6	0.0
1.426	1.923	26.3	1	0.0	0.0	0.0	0.0
1.497	1.906	26.5	1	0.0	0.0	0.0	0.0
1.568	1.91	26.4	1	0.0	0.0	0.0	0.0
1.639	1.916	26.4	1	0.0	0.0	0.0	0.0

Exhibit 9A. One stop, loop by loop. Floor test, step 4, last run, 15 Oct 2026, avoid.py 80 rpm, stop at 30 cm, data in avoid_80_30.csv. Duty drops to 0 in the first loop that reads 30 cm or less (bold). The wheels coast to 0 in 0.38 s while the sensor sees the wall close from 29.5 to 26.3 cm. Speed was 70 rpm, still rising towards 80 from a close start. Loops average 82 ms. rpm is measured over the loop before each row.